

EXPERIMENT: 27

IMPLEMENTATION OF A DNS SERVER AND CLIENT IN JAVA/C USING UDP SOCKET

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>
```

```
int main()
{
    int server, client;
    struct sockaddr_in serverAddr, clientAddr;
    socklen_t len;
    char domain[100];
    char ip[100];

    char *hosts[] = {
        "google.com",
        "yahoo.com",
        "facebook.com",
        "amazon.com"
    };

    char *addresses[] = {
        "142.250.183.14",
        "98.137.11.163",
        "157.240.241.35",
        "205.251.242.103"
```

```
};
```

```
int n = 4;
```

```
int i, found;
```

```
/* Create UDP server socket */
```

```
server = socket(AF_INET, SOCK_DGRAM, 0);
```

```
serverAddr.sin_family = AF_INET;
```

```
serverAddr.sin_port = htons(8080);
```

```
serverAddr.sin_addr.s_addr = INADDR_ANY;
```

```
bind(server, (struct sockaddr *)&serverAddr,  
      sizeof(serverAddr));
```

```
printf("DNS Server started...\n");
```

```
/* Client socket */
```

```
client = socket(AF_INET, SOCK_DGRAM, 0);
```

```
clientAddr.sin_family = AF_INET;
```

```
clientAddr.sin_port = htons(8080);
```

```
clientAddr.sin_addr.s_addr = inet_addr("127.0.0.1");
```

```
printf("Enter domain name: ");
```

```
scanf("%s", domain);
```

```
/* Send domain name to server */
```

```
sendto(client, domain, strlen(domain) + 1, 0,
```

```

        (struct sockaddr *)&clientAddr,
        sizeof(clientAddr));

/* Server receives request */
len = sizeof(clientAddr);

recvfrom(server, domain, sizeof(domain), 0,
        (struct sockaddr *)&clientAddr, &len);

found = 0;

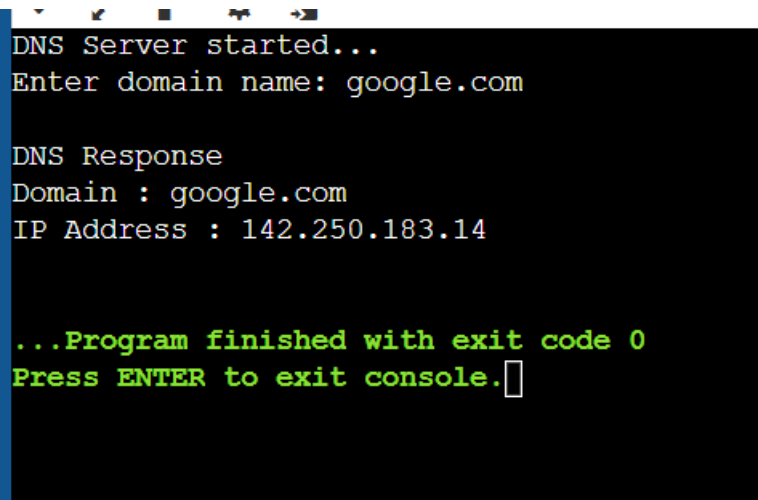
/* Search domain */
for (i = 0; i < n; i++)
{
    if (strcmp(domain, hosts[i]) == 0)
    {
        strcpy(ip, addresses[i]);
        found = 1;
        break;
    }
}

if (found == 0)
    strcpy(ip, "Domain does not exist");

/* Send IP address to client */
sendto(server, ip, strlen(ip) + 1, 0,
        (struct sockaddr *)&clientAddr,
        len);

```

```
/* Client receives response */  
recvfrom(client, ip, sizeof(ip), 0, NULL, NULL);  
  
printf("\nDNS Response\n");  
printf("Domain : %s\n", domain);  
printf("IP Address : %s\n", ip);  
  
close(server);  
close(client);  
  
return 0;  
}
```



The screenshot shows a terminal window with a black background and white text. At the top, it says "DNS Server started...". Below that, it prompts "Enter domain name: google.com". The next line is "DNS Response", followed by "Domain : google.com" and "IP Address : 142.250.183.14". At the bottom, it says "...Program finished with exit code 0" and "Press ENTER to exit console." with a cursor.

```
DNS Server started...  
Enter domain name: google.com  
  
DNS Response  
Domain : google.com  
IP Address : 142.250.183.14  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```